

Immunogenicity, safety, and efficacy of the vaccine H56:IC31 in reducing the rate of tuberculosis disease recurrence in HIV-negative adults successfully treated for drug-susceptible pulmonary tuberculosis: a double-blind, randomised, placebo-controlled, phase 2b trial



Álvaro H Borges, Marisa Russell, Dereck Tait, Thomas J Scriba, Elisa Nemes, Per Skallerup, Elana van Brakel, Andrea M Cabibbe, Daniela M Cirillo, Mildie Leuvennink-Steyn, Kathryn T Rutkowski, Grith K Wood, Birgit Thierry-Carstensen, Pernille N Tingskov, Emilie C Meldgaard, Max P Kristiansen, Rie E Søndergaard, Christian H Hansen, Frank Follmann, Charlotte G Jensen, Anele Gela, Nyanda E Ntinginya, Morten Ruhwald, Justin Shenje, Lisa White, Craig Innes, Pearl Selepe, Beatrice Ngaraguza, Chantelle Holmgren, Tarryn Collings, Peter Andersen, Rodney Dawson, Gavin Churchyard, Issa Sabi, Andreas H Diacon, Rasmus Mortensen*, Mark Hatherill*, on behalf of the POR TB study group

Summary

Background People with tuberculosis who complete treatment remain at risk of recurrent disease. The vaccine H56:IC31 has been shown to be safe and immunogenic in phase 1 and 2 studies, but whether it can reduce the risk of tuberculosis recurrence is unknown.

Methods In a double-blind, randomised, placebo-controlled, phase 2b trial in South Africa (five clinical trial sites) and Tanzania (one clinical trial site), we enrolled participants aged 18–60 years, without HIV, who had completed more than 5 months (22 weeks) of treatment for drug-susceptible pulmonary tuberculosis. During trial screening (≤ 7 days after starting treatment), two sputum samples were obtained and frozen for later comparison to recurrent isolates by whole-genome sequencing (WGS). Eligible participants were randomly assigned (1:1; block size of four) to receive two intramuscular doses in the deltoid, 56 days apart, of H56:IC31 or placebo. After the first dose of H56:IC31 or placebo, participants were followed up until study day 421 (1 year after the second dose) and checked at each visit for tuberculosis signs and symptoms. If tuberculosis was suspected, two sputum samples were obtained: one sample was tested by automated molecular test (Xpert MTB/RIF Ultra) and sent for liquid culture; and the other sample was stored frozen for later analysis by whole-genome sequencing (WGS). At the last visit (day 421), two sputum samples were obtained from all sputum-productive participants, regardless of symptoms, to detect cases of asymptomatic tuberculosis. The primary endpoint was culture-confirmed recurrent pulmonary tuberculosis (due to relapse with the same strain, reinfection by a different strain, or indeterminate) occurring during the period starting at day 70 (14 days after the second dose) and ending on day 421 (1 year after the second dose). Vaccine efficacy against recurrent tuberculosis was derived from Cox proportional hazards models. Secondary endpoints included vaccine efficacy to prevent tuberculosis relapse or reinfection independently, as differentiated by WGS, and safety and immunogenicity outcomes (H56-specific CD4 T-cell responses and humoral anti-H56 IgG responses). Primary analysis of vaccine efficacy was based on modified intention-to-treat (mITT), in all randomly assigned participants except those with tuberculosis disease recurrence or who withdrew before day 70 (or 14 days after the second dose for those who received both doses). Safety was assessed in all randomly assigned participants who received at least one dose of vaccine or placebo. The trial was registered with ClinicalTrials.gov, NCT03512249, and is complete.

Findings 831 participants (mean age 34·7 years [SD 11·1]; 229 [28%] female and 602 [72%] male; 549 [66%] Black) were enrolled from Jan 31, 2019, to Jan 20, 2022; 415 participants were randomly assigned to receive H56:IC31 and 416 to receive placebo. Follow-up was completed by March 20, 2023 (mean follow-up duration 410·1 days [SD 82·8]). In the primary mITT analysis, recurrent tuberculosis occurred in 23 of 400 participants in the H56:IC31 group (12 relapses, eight reinfections, and three indeterminate); and in 14 of 406 in the placebo group (six relapses, seven reinfections, and one indeterminate). Vaccine efficacy for prevention of recurrence was –73·8% (95% CI –246·9 to 9·8; $p=0\cdot10$). Vaccine efficacy for prevention of relapse was –116·1% (–522·2 to 16·3; $p=0\cdot11$) and for prevention of reinfection was –21·1% (–245·3 to 56·5; $p=0\cdot71$). 2 weeks after the planned second dose, H56:IC31 had significantly increased the frequencies of H56-specific CD4 T cells expressing interferon- γ , tumour necrosis factor, interleukin (IL)-2, or IL-17 in vaccinees (median percentage of CD4 T cells, 0·35% [IQR 0·19 to 0·57]) compared with placebo (0·11% [0·09 to 0·23]; $p<0\cdot0001$). H56-specific IgG responses were significantly higher in H56:IC31 recipients (median arbitrary units per mL, 6·84 [IQR 1·64 to 32·8]) than in placebo recipients (1·94 [1·05 to 3·86]; $p<0\cdot0001$). A greater proportion of H56:IC31 recipients had mild-to-moderate injection site reactions than placebo recipients (165 [40%] of 415 vs

Lancet Infect Dis 2025; 25: 751–63

Published Online
March 5, 2025
[https://doi.org/10.1016/S1473-3099\(24\)00814-4](https://doi.org/10.1016/S1473-3099(24)00814-4)

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*Contributed equally, joint last authors

Department of Infectious Disease Immunology (Á H Borges PhD, F Follmann PhD, R Mortensen PhD) and **Department of Vaccine Development** (P Skallerup PhD, G K Wood MSc, B Thierry-Carstensen MSc, P N Tingskov RN, E C Meldgaard MSc, M P Kristiansen MSc, C G Jensen PhD), **Centre for Vaccine Research, Statens Serum Institut, Copenhagen, Denmark**; IAVI, Cape Town, South Africa (M Russell BSc, E van Brakel MD, M Leuvennink-Steyn BSc, K T Rutkowski MS); Cape Town, South Africa (D Tait MBChB); **South African Tuberculosis Vaccine Initiative, Institute of Infectious Disease and Molecular Medicine and Division of Immunology, Department of Pathology, University of Cape Town, Cape Town, South Africa** (Prof T J Scriba PhD, E Nemes PhD, A Gela PhD, J Shenje MBChB, Prof M Hatherill MD); **Emerging Bacterial Pathogens Unit, IRCCS Ospedale San Raffaele, Milano, Italy** (A M Cabibbe PhD, Prof D M Cirillo PhD); **BioStata, Birkerød, Denmark** (R E Søndergaard MSc);

Department of Infectious Disease Epidemiology and Prevention, Statens Serum Institut, Copenhagen, Denmark (C H Hansen PhD); MRC International Statistics and Epidemiology Group, London School of Hygiene & Tropical Medicine, London, UK (C H Hansen); Mbeya Medical Research Center, National Institute for Medical Research, Mbeya, Tanzania (N E Ntinginya PhD, B Ngaruza MD, I Sabi PhD); Foundation for Innovative New Diagnostics, Geneva, Switzerland (M Ruhwald PhD); Centre for Tuberculosis Research Innovation, University of Cape Town Lung Institute, Cape Town, South Africa (L White MD, Prof R Dawson MD); The Aurum Institute, Johannesburg, South Africa (C Innes MSc, P Selepe MD, Prof G Churchyard PhD); TASK, Cape Town, South Africa (C Holmgren MD, T Collings MD, Prof A H Diacon PhD); Novo Nordisk Foundation, Hellerup, Denmark (Prof P Andersen DMSc); Department of Immunology and Microbiology, University of Copenhagen, Copenhagen, Denmark (Prof P Andersen); School of Public Health, Faculty of Health Sciences, University of Witwatersrand, Johannesburg, South Africa (Prof G Churchyard); Department of Medicine, Vanderbilt University, Nashville, TN, USA (Prof G Churchyard)

Correspondence to: Dr Álvaro H Borges, Department of Infectious Disease Immunology, Centre for Vaccine Research, Statens Serum Institut, 2300 KBH S, Copenhagen, Denmark
albs@ssi.dk

78 [19%] of 416). No treatment-related serious adverse events were reported. Two participants who received H56:IC31 and six who received placebo died.

Interpretation Vaccination with H56:IC31 at treatment completion for pulmonary tuberculosis did not reduce the risk of recurrent disease. H56:IC31 was well tolerated and immunogenic but might have increased the risk of relapses by endogenous strains.

Funding The European and Developing Countries Clinical Trials Partnership (EDCTP2) supported by the EU (grant number RIA2016V-1631, POR TB consortium). Additional funding to support completion of the trial was provided by the Statens Serum Institut, Aurum Institute, and the South African Tuberculosis Vaccine Initiative.

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Introduction

In 2023, 10·8 million people developed tuberculosis disease worldwide, with an estimated 1·25 million deaths.¹

Bacille Calmette-Guérin (BCG), the only licensed tuberculosis vaccine for more than 100 years, confers partial protection against severe disease in childhood,

Research in context

Evidence before this study

We searched PubMed on Aug 1, 2024, using the terms “H56:IC31” and “clinical trial” and “prevention of tuberculosis recurrence”, with no restrictions on publication dates or language. We found that H56:IC31 had an acceptable safety profile and was immunogenic in five randomised clinical trials: (1) the first-in-human trial in 25 individuals (NCT01967134); (2) a dose-finding trial in 98 individuals (NCT01865487); (3) a phase 1b trial with 24 adolescents (NCT02378207); (4) a phase 1 trial with 16 participants vaccinated at the end of treatment for drug-sensitive tuberculosis in South Africa (NCT02375698); and (5) a phase 1/2 trial with 22 participants vaccinated after 3 months of treatment for drug-sensitive tuberculosis in Norway (NCT02503839). The first three trials were conducted among *Mycobacterium tuberculosis*-infected and *M tuberculosis*-uninfected, BCG-vaccinated, HIV-negative individuals in South Africa.

Added value of this study

The current paper is, to our knowledge, the first report of a randomised, placebo-controlled clinical trial to assess the efficacy of a tuberculosis vaccine candidate administered at the end of tuberculosis treatment to prevent disease recurrence, assessed over a 1-year follow-up. Whole-genome sequencing or targeted next-generation sequencing of paired sputum samples was used to differentiate relapses by the same bacterial strain from reinfections with a different strain. Because the incidence of recurrent tuberculosis disease following completion of tuberculosis treatment is several-times higher than the incidence of first-episode tuberculosis, promising tuberculosis vaccine candidates can be assessed in proof-of-concept efficacy trials with a prevention-of-recurrence (POR) design. Given the high number of clinical endpoints in POR trials such trials can be designed smaller, shorter, and more cost-effective than trials involving the general population. In this study, we found that vaccination with H56:IC31 after 5 months of tuberculosis treatment was safe, well

tolerated, and immunogenic, but did not reduce the risk of the primary endpoint, recurrent tuberculosis, in the year following last vaccination. Counterintuitively, we observed an excess of nine cases of recurrence (six of which were cases of relapse with the same bacterial strain, as a trial secondary endpoint) among H56:IC31 vaccinees compared with placebo recipients in our study.

Implications of all the available evidence

POR trials might set a high performance bar for testing new tuberculosis vaccine candidates because immune mechanisms required to control mycobacteria, which are perturbed during active pulmonary disease, are not fully restored after successful treatment and might not be appropriately modulated by vaccination. There are several possible reasons for the potentially increased risk for relapse in the H56:IC31 group, one of these reasons being that the trial was not powered to assess vaccine efficacy against this secondary endpoint. Additionally, despite randomisation, there might have been unknown imbalances in disease severity at baseline, treatment adherence, or sputum smear-negative, but culture-positive, disease, towards the end of treatment, which in turn would have biased the risk for relapse by study group. Alternatively, vaccine-induced immune responses could have promoted the development of relapse by reactivation of endogenous bacteria. The results of this study, although not statistically significant, raise the question as to whether participants treated for active tuberculosis can be protected against relapse by vaccination. The results of an ongoing trial in India and Bangladesh testing the recombinant Bacille Calmette-Guérin vaccine VPM1002 (NCT03152903) with a similar POR design will be helpful to clarify this. Our results cannot be generalised to trials of therapeutic vaccination, in which participants are vaccinated while receiving standard tuberculosis therapy. In this regard, a planned therapeutic trial testing the tuberculosis vaccine ID93 + GLA-SE during tuberculosis treatment (NCT06205589) should be particularly informative.

but not in adults.² Without an effective vaccine, the targets for reducing tuberculosis burden set out in the WHO End Tuberculosis Strategy remain elusive.³ A positive efficacy signal of the subunit vaccine M72/AS01_E against progression from infection to pulmonary disease^{4,5} has given momentum to the search for a better tuberculosis vaccine.

The incidence of recurrent tuberculosis, defined as a new diagnosis of tuberculosis after a successfully treated tuberculosis episode, is several-times higher than the incidence of new tuberculosis disease in the general population in endemic regions.^{6–9} Vaccination of patients with tuberculosis during or on completion of tuberculosis treatment might boost protective immune responses and enhance control or killing of persistent mycobacteria, which are thought to remain viable after supposed cure.¹⁰ Thus, this target population might benefit from an effective vaccine that increases the rate of non-relapsing cure; and tuberculosis vaccine candidates could be assessed in small proof-of-concept efficacy trials with a prevention-of-recurrence (POR) design.¹¹

The vaccine candidate H56:IC31 has provided protection against tuberculosis in animal models^{12,13} and shown acceptable safety and immunogenicity in uninfected adolescents,¹⁴ infected and uninfected adults,^{15,16} and individuals with tuberculosis disease during¹⁷ and on completing¹⁸ tuberculosis treatment. In the present trial by the POR TB study group, we investigated whether H56:IC31 vaccination upon treatment completion could reduce the subsequent risk of recurrent tuberculosis. We also used whole-genome sequencing (WGS) of paired sputum samples to differentiate relapses by the same bacterial strain from reinfections by a different strain.

Methods

Study design

We did a double-blind, randomised, placebo-controlled, phase 2b, clinical endpoint-driven trial at six clinical trial sites, in South Africa (five sites: TASK, South African Tuberculosis Vaccine Initiative [SATVI], and University of Cape Town Lung Institute [UCTLI] sites in Cape Town, and The Aurum Institute sites in Tembisa and Klerksdorp) and Tanzania (one site: National Institute for Medical Research [NIMR], Mbeya).

The trial was done in accordance with the Declaration of Helsinki, the International Council for Harmonisation Guideline for Good Clinical Practice (version E6, revision 2), and national Good Clinical Practice guidelines, and is registered with ClinicalTrials.gov, NCT03512249. The study protocol was approved by the South Africa Health Products Regulatory Authority (Pretoria, South Africa; approval number 20180624), by the Tanzania Medicines and Medical Devices Authority (Dar es Salaam, Tanzania; approval number TFDA0019/CTR/00147/18), and by all relevant institutional ethics committees in South Africa and Tanzania.

Participants

Participants were recruited from clinics at the start of treatment for tuberculosis disease. We enrolled male and female participants aged 18–60 years, without HIV, with drug-susceptible pulmonary tuberculosis, who had completed more than 5 months (22 weeks) of tuberculosis treatment, with treatment ongoing at the time of the first trial dose (vaccine or placebo), and with a total treatment time not extending beyond 28 weeks. Study participants had been receiving treatment for drug-susceptible pulmonary tuberculosis with standard first-line treatment that included an initial phase of 2 months of isoniazid, rifampicin, pyrazinamide, and ethambutol, followed by a continuation phase of 4 months of isoniazid and rifampicin. Trial vaccination was initiated while tuberculosis treatment was ongoing, with a second dose (vaccine or placebo) administered after treatment completion, to induce adaptive immune responses that could prevent bacterial regrowth in the absence of antibiotic pressure.¹⁸ Participants needed a confirmed smear-negative sputum sample after 22 weeks of treatment (per national tuberculosis control programmes in South Africa and Tanzania, as the definition of cure), although asymptomatic participants unable to produce sputum were also eligible for inclusion. All participants gave written informed consent before entering the trial and random assignment, and agreed to stay in contact with the trial site for the duration of the study and to give access to their medical records for study-related purposes. Female participants of childbearing potential were required to have a negative pregnancy test (at screening, enrollment, and before the second dose) and to agree to use acceptable contraceptive measures (such as a sterile sexual partner, sexual abstinence, hormonal contraceptives, or an intrauterine device) during the trial.

Screened individuals were excluded if they had a diagnosis or co-diagnosis of extrapulmonary tuberculosis, were hospitalised for the current episode of drug-susceptible pulmonary tuberculosis disease, or had a history of receipt of treatment against active tuberculosis, before the current treatment episode, within the past 5 years. Screened individuals were also excluded if they had a history of or ongoing severe disease that, in the opinion of the investigator, might affect the safety of the participant or the immunogenicity of the investigational product, including type 1 diabetes, allergic disease or reactions likely to be exacerbated by any component of the investigational product, laboratory evidence of immunodeficiency, autoimmune disease or immunosuppression, chronic hepatitis, severe anaemia (defined as haemoglobin <10 g/dL or a haematocrit value of <30% based on the most recent hematology tests obtained before randomisation), or a BMI lower than 13 kg/m².

The prespecified inclusion and exclusion criteria are fully described in the trial protocol (see protocol version 5.0 sections 3.2.3–3.2.4 in appendix 1).

See Online for appendix 1

Randomisation and masking

Eligible participants were randomly assigned (1:1) to H56:IC31 or placebo. Randomisation was done with a randomly generated sequence of numbers managed by a validated module in the electronic case report form. The randomisation list was prepared by an unmasked statistician who had no further involvement in the trial. We used block randomisation with a block size of four. The day of randomisation and administration of the first vaccine or placebo dose for each participant was defined as study day 0. H56:IC31 and placebo were prepared by an unmasked study vaccine manager in a unit-dose syringe labelled with a participant identification number. The study vaccine manager was not involved in the rest of the trial. Syringes for H56:IC31 and placebo were identical in appearance and the contents were masked by a translucent label. All other study staff involved in the administration of vaccine or placebo injection, and follow-up for clinical and laboratory assessments, were masked to treatment allocation. All participants were masked to treatment allocation.

Procedures

Randomly assigned participants received two intramuscular doses in the deltoid, 56 days apart, of either H56:IC31 or placebo. The H56:IC31 vaccine was manufactured at Statens Serum Institut (Copenhagen, Denmark) according to good manufacturing practice of the European Medicines Agency. H56:IC31 has two components: H56, a recombinant fusion protein of antigens Ag85B, ESAT-6, and Rv2660c; and IC31, an oligonucleotide-based adjuvant comprising the peptide KLKL₅KLK and the oligonucleotide ODN1a. IC31 was developed by Valneva (Vienna, Austria) and licensed to Statens Serum Institut. The vaccine was supplied as a combined dose at a concentration of 5 µg per 0.5 mL buffer of H56 with IC31 at a concentration of 500 nmol KLKL₅KLK and 20 nmol ODN1a per 0.5 mL buffer. 0.9% saline was used as placebo. Two batches of H56:IC31 were used in the trial with expiry dates of July 1, 2020 (Statens Serum Institut batch number 521202) and March 20, 2022 (batch number 711101).

Study participants had two screening visits during tuberculosis treatment. At the first screening visit (≤7 days after starting treatment), two separate sputum samples were obtained and frozen (at −80°C) for later comparison to recurrent isolates by WGS. The second screening visit to confirm sputum smear negativity was done more than 5 months (22 weeks) after treatment started. From visit 4 (study day 14) and onwards, tuberculosis symptoms screening was done at all study visits and by five telephone calls. The immunisation timelines, tests, and study procedures done at each study visit are summarised in appendix 2 (pp 1–4). Information on sex and race was reported by site investigators as self-reported by participants.

The trial conduct was delayed by COVID-19 pandemic measures in South Africa and Tanzania. A protocol amendment was implemented to allow vaccination of trial participants with SARS-CoV-2 vaccines recommended by national vaccination programmes (if given more than 28 days before or after the time of administration of clinical trial product). A second protocol amendment to complete recruitment before study vaccine expiry allowed screening visits to occur concurrently after 5 months of treatment. All versions of the protocol (1.0–5.0) are provided in appendix 1. All minor and major protocol changes are described in detail in protocol version 5.0 (sections 12.1–12.4).

After the first dose of H56:IC31 or placebo, participants were followed up until study day 421 (visit 8; 1 year after the second dose) and checked at every study visit and telephone call for tuberculosis signs and symptoms (unexplained cough for more than 2 weeks, fever, night sweats, loss of weight, or haemoptysis). If tuberculosis was suspected from visit 4 (day 14) onwards, two sputum samples were obtained: one sample was tested by automated molecular test (Xpert MTB/RIF Ultra, Cepheid, Sunnyvale, USA; appendix 2 p 5) and sent for liquid culture; the other sample was stored frozen (at −80°C) for later analysis by WGS with a paired-end Nextera XT DNA Library Preparation Kit (Illumina, San Diego, USA) (appendix 2 p 5). As a back-up option to WGS, targeted next-generation sequencing (tNGS) was done with a Deeplex MycTB assay (GenoScreen, Lille, France; appendix 2 p 5). The WGS procedures have been described elsewhere.¹⁹ tNGS procedures are described in appendix 2 (p 5). At the last visit on day 421, two sputum samples were obtained from all participants to detect cases of asymptomatic tuberculosis. Asymptomatic, sputum-unproductive participants were deemed to be free from recurrent tuberculosis.

Among recurrent cases confirmed by culture, isolates from paired (screening and post-dosing) sputum samples differing by less than or equal to five single nucleotide polymorphisms or alleles were considered relapses by the same bacterial strain;²⁰ otherwise, they were considered reinfections by different strains. These assessments were made by WGS or tNGS.

H56-specific CD4 T cells were measured by a whole-blood intracellular cytokine staining assay (appendix 2 pp 5–9) and anti-H56 IgG concentrations were quantified by ELISA (appendix 2 p 9) at day 0 (visit 3) and day 70 (visit 6) in the first 100 participants randomly assigned overall at the SATVI site in South Africa and the NIMR site in Tanzania (a protocol-specified immunogenicity cohort; 46 participants in the H56:IC31 group and 54 in the placebo group).

Solicited injection site reactions, solicited systemic adverse events, and unsolicited adverse events were collected for 14 days after the first and second doses, and any serious adverse events (defined as events that resulted in death, were life-threatening, required

hospitalisation or prolongation of existing hospitalisation, resulted in significant or persistent disability or incapacity, or congenital anomaly or birth defect, or any other medically important condition) were collected after the first dose until the study end. All of these events were considered treatment-emergent adverse events. Adverse events were classified according to the Medical Dictionary for Regulatory Activities (version 21.1). Treatment-related adverse events were defined as adverse events for which there was a reasonable possibility that the investigational product contributed to the adverse events. This included adverse events that were considered possibly, probably, or certainly related. The intensities of adverse events were defined as mild (noticeable but had no impact on normal activities; did or did not require over-the-counter treatment managed by the participant), moderate (had some impact on normal activities and might have required general symptomatic medical intervention by a health-care professional or by the participant), or severe (might have been incapacitating, leading to suspension of normal daily activities, and would generally have required more immediate medical evaluation and intervention by a health-care professional). Solicited adverse events were collected by diary. Serum chemistry, haematology, and urinalysis safety tests were done for all participants at screening (visit 2). These safety tests were repeated at day 14 (visit 4) and day 70 (visit 6) for participants in a protocol-specified safety cohort (ie, the first 150 randomly assigned participants in the trial; 76 participants in the H56:IC31 group and 74 in the placebo group). A safety test could also be done at the discretion of the investigator for all participants at any visit, or at an early termination visit, if applicable. An internal data review committee convened twice during the study to review safety data in a masked manner. No interim analysis was done.

Outcomes

The primary endpoint was recurrent pulmonary tuberculosis (due to relapse, reinfection, or indeterminate), defined as culture-confirmed pulmonary tuberculosis occurring during the period starting at day 70 (14 days after the second dose) and ending at day 421 (1 year after the second dose). The efficacy of H56:IC31 compared with placebo in reducing the rates of relapse or reinfection was assessed as secondary endpoints.

Secondary immunogenicity endpoints were H56-specific CD4 T cell-mediated immune responses, measured as the frequencies of all CD4 T cells that were H56-specific CD4 T cells expressing any combination of interferon (IFN)- γ , tumour necrosis factor (TNF), interleukin (IL)-2, or IL-17, and humoral anti-H56 IgG responses, measured as the concentrations of anti-H56 IgG. Secondary safety endpoints were treatment-emergent solicited injection site reactions, solicited systemic adverse events, unsolicited adverse events, and any serious adverse events.

Prespecified exploratory endpoints included the rate of tuberculosis disease recurrence (relapse or reinfection or indeterminate), defined as participants meeting the primary endpoint of tuberculosis disease recurrence and participants who started new tuberculosis treatment without confirmation of *Mycobacterium tuberculosis* by culture of sputum; the rate of tuberculosis disease recurrence (relapse or reinfection or indeterminate), defined as participants meeting the primary endpoint of tuberculosis disease recurrence based on confirmation of *M tuberculosis* by Xpert MTB/RIF Ultra test or culture of sputum; and the rate of tuberculosis disease recurrence (relapse or reinfection or indeterminate), defined as participants meeting the primary endpoint of tuberculosis disease recurrence and participants diagnosed between 30 days after the first dose and 14 days after the second dose, based on confirmation of *M tuberculosis* by culture of sputum.

Statistical analyses

The trial was designed to provide 80% power to detect a vaccine efficacy of 60% against recurrent tuberculosis in the H56:IC31 group compared with placebo at a one-sided significance level of 0.1, equivalent to a two-sided significance level of 0.2. The lenient significance level was chosen as even weak evidence of a vaccine effect would merit further research into the vaccine candidate. The trial was not powered to detect vaccine efficacy against either relapse or reinfection independently. This design required that a minimum of 23 endpoints accrued across both groups. Assuming a rate of recurrent tuberculosis after the second dose (from day 70) of 4% per year in the placebo group,^{9,21} a trial duration of 1 year after the second dose and a loss-to-follow up rate of 10%, we estimated a total sample size of 900.

All analysis sets were planned a priori. The intention-to-treat (ITT) analysis set consisted of all randomly assigned participants. The safety analysis set consisted of all randomly assigned participants who received at least one dose of trial vaccine or placebo. The modified ITT (mITT) analysis set included all randomly assigned participants except those with tuberculosis disease recurrence before visit 6 on day 70 (or 14 days after the second dose for those who received both doses). The per-protocol analysis set consisted of all participants who received both doses of H56:IC31 or placebo within the specified dose intervals, who entered the evaluation period for efficacy 14 days after receipt of the second dose with no HIV seroconversion, and who had no major protocol deviation of clinical or statistical significance. Participants with a recurrence of tuberculosis exactly on day 70 (or 14 days after the second dose) were excluded from the mITT and per-protocol analysis sets. The immunogenicity analysis set consisted of participants from the mITT population who received both doses of H56:IC31 or placebo and who were included in the

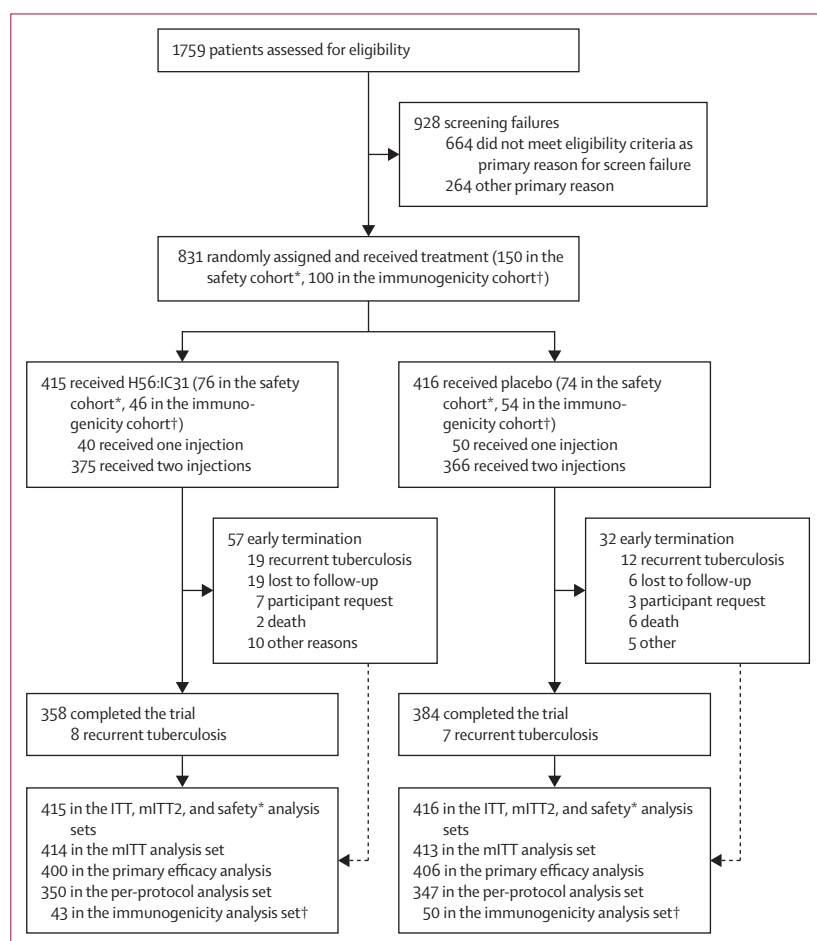


Figure 1: CONSORT diagram

As per trial protocol, the primary efficacy analysis was based on the mITT set. 46 participants either discontinued the trial early due to recurrent tuberculosis (31 participants: 19 in the H56:IC31 group and 12 in the placebo group) or completed the trial with recurrent tuberculosis (15 participants: eight in the H56:IC31 group and seven in the placebo group). 41 of these participants had recurrent tuberculosis according to the primary efficacy endpoint definition (ie, pulmonary tuberculosis confirmed by culture). Of the 41 participants with recurrent tuberculosis confirmed by culture, 37 participants (23 in the H56:IC31 group and 14 in the placebo group) were included in the mITT population, which was the analysis set for the primary efficacy analysis. The four participants who were excluded from the analysis (one in the H56:IC31 group and three in the placebo group) had onset of recurrent tuberculosis before visit 6 on day 70 (ie, before 14 days after the second injection), which was protocol-defined exclusion criteria for the mITT population. In addition, 21 participants (14 in the H56:IC31 group and seven in the placebo group) included in the mITT set did not contribute to the primary efficacy analysis, as they withdrew from trial before reaching day 70 for reasons other than recurrent tuberculosis. ITT=intention to treat. mITT=modified intention to treat. *The safety cohort comprised the first 150 randomly assigned participants, whereas the safety analysis set comprised all participants who received at least one dose of trial vaccine or placebo. †The immunogenicity cohort comprised the first 100 participants randomly assigned overall at the SATVI site in South Africa and the NIMR site in Tanzania, and the immunogenicity analysis set comprised participants in the mITT population who received both doses of H56:IC31 or placebo and who were included in the immunogenicity cohort.

immunogenicity cohort. We also conducted analyses using a second mITT (mITT2) analysis set, which included all randomly assigned participants except those with tuberculosis disease recurrence before 30 days after the first dose.

Categorical data were summarised by study group as counts and percentages of participants. For calculation of percentages, the denominator was the number of

participants in the analysis set. Continuous data were presented as means and SDs if normally distributed, or as medians and IQRs in case of a skewed distribution.

As per study protocol, the primary analysis of vaccine efficacy against recurrent tuberculosis was based on the mITT analysis set. This analysis included all randomly assigned participants who did not develop recurrent tuberculosis or withdraw from the study before day 70, in order to exclude cases of early treatment failure and allow time for the vaccine to induce protective immune responses. Tuberculosis recurrence risk was calculated as the rate of recurrence per 100 person-years of follow-up, with 95% CIs estimated using the Poisson distribution. Time-to-event methods, including Kaplan–Meier survival curves with origin time at day 70 stratified by study site, and Cox proportional hazards regression models with treatment and trial site as fixed effects, were used to estimate vaccine efficacy, calculated as 1 minus the hazard ratio with 95% CIs. Treatment group inference was evaluated with a two-sided likelihood ratio test. No adjustment for confounding covariates was made. For the primary efficacy endpoint, missing values were handled by censoring. Participants who did not have tuberculosis recurrence confirmed by culture of sputum samples and asymptomatic participants not capable of producing sputum were censored at the time of death, withdrawal, last contact, or 1 year after day 70, whichever occurred first. No imputation of missing values was done. Two-sided p values were obtained. Secondary and exploratory efficacy endpoints were analysed with the same methods as for the primary efficacy endpoint of tuberculosis recurrence. To support robustness and consistency of the results, we also did prespecified analyses of vaccine efficacy endpoints in the per-protocol, ITT, and mITT2 sets.

To further assess consistency of the results for the primary endpoint, subgroups were selected a priori, defined by the following factors: clinical trial site, country, age, sex, smoking status, anaemia, comorbidities (defined as a history of chronic or intermittent diseases, medical conditions, or symptoms judged to influence the primary endpoint as assessed by the medical expert (AHB) of the trial before unmasking, including type 2 diabetes), BMI, steroid or anti-inflammatory drug use during the trial, number of doses of H56:IC31 or placebo (one vs two), and COVID-19-related protocol deviations (ie, protocol deviations occurring due to restrictions during COVID-19 lockdowns), and included in expanded Cox models with interaction terms for the subgroup variable and study intervention. Trial site, country, age, sex, smoking status, BMI, comorbidities, and anaemia were defined at baseline. Subgroup analyses were done in the primary mITT analysis population.

Immunogenicity was assessed in the immunogenicity analysis set at baseline (day 0) and 2 weeks after the planned second dose (day 70), excluding participants with missing data at either timepoint. Differences in

responses between day 0 and day 70, and between placebo and H56:IC31 recipients, were compared with Wilcoxon signed-rank test and Mann-Whitney test, respectively. Assessments of safety endpoints were based on the safety analysis set.

The complete statistical analysis plan for the trial data is available in appendix 3. Statistical analyses were done with SAS software (version 9.4). In all analyses, a p value of less than 0.05 was considered to indicate statistical significance.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

From Jan 31, 2019 to Jan 20, 2022, 831 participants were enrolled and randomly assigned to receive H56:IC31 (n=415) or placebo (n=416; figure 1). Enrolment was delayed due to the COVID-19 pandemic and was stopped before the target sample size of 900 was reached, due to the expiry date of study vaccine. Overall, the 831 enrolled participants had a mean age of 34.7 years (SD 11.1), 229 (28%) were female and 602 (72%) were male, 549 (66%) were Black, 633 (76%) were recruited in South Africa, 385 (46%) were current smokers, and 234 (28%) had comorbidities. We found no notable between-group differences in baseline characteristics (table 1). Follow-up was completed by March 20, 2023, with a mean duration of follow-up of 410.1 days (SD 82.8).

All randomly assigned participants received at least one dose of the allocated intervention. 400 of 415 participants who received H56:IC31 and 406 of 416 who received placebo were included in the primary mITT analysis (figure 1). 350 participants in the H56:IC31 group and 347 participants in the placebo group were included in per-protocol analyses. 40 participants in the H56:IC31 group and 50 participants in the placebo group did not receive a second dose and were excluded from per-protocol analyses. For the remaining participants excluded from the per-protocol population, the most common reasons were tuberculosis treatment errors and missing Xpert MTB/RIF Ultra results at visit 8. Major protocol deviations and protocol deviations that led to the exclusion of individuals from per-protocol analyses are listed in appendix 2 (p 10).

In the primary mITT analyses, recurrent tuberculosis occurred in 23 of 400 participants in the H56:IC31 group (12 relapses, eight reinfections, and three indeterminate); and in 14 of 406 participants in the placebo group (six relapses, seven reinfections, and one indeterminate). These numbers were equivalent to a tuberculosis recurrence risk per 100 person-years of follow-up of 6.3% (95% CI 4.0 to 9.4) and 3.7% (2.0 to 6.2) in the two groups, respectively. Vaccine efficacy for prevention of recurrence was calculated as -73.8% (95% CI

246.9 to 9.8; p=0.10), indicating a possible increase in the risk of recurrence in the vaccinated group, although not statistically significant (figure 2A; appendix 2 p 11). For secondary efficacy endpoints, vaccine efficacy for prevention of relapse was -116.1% (-522.2 to 16.3; p=0.11) and for prevention of reinfection was -21.1% (-245.3 to 56.5; p=0.71; figure 2B–C).

See Online for appendix 3

	H56:IC31 group (N=415)	Placebo group (N=416)	Total (N=831)
Mean age (SD), years	34.3 (11.2)	35.1 (11.0)	34.7 (11.1)
Sex			
Female	112 (27%)	117 (28%)	229 (28%)
Male	303 (73%)	299 (72%)	602 (72%)
Race*			
Black	276 (67%)	273 (66%)	549 (66%)
Mixed Cape ancestry	132 (32%)	142 (34%)	274 (33%)
Other	7 (2%)	1 (<1%)	8 (1%)
Country			
South Africa	323 (78%)	310 (75%)	633 (76%)
Tanzania	92 (22%)	106 (25%)	198 (24%)
Trial site			
SATVI	47 (11%)	61 (15%)	108 (13%)
UCTLI	69 (17%)	70 (17%)	139 (17%)
TASK	128 (31%)	101 (24%)	229 (28%)
Aurum Klerksdorp	46 (11%)	45 (11%)	91 (11%)
NIMR Mbeya	92 (22%)	106 (25%)	198 (24%)
Aurum Tembisa	33 (8%)	33 (8%)	66 (8%)
Smoking status			
Non-smoker	149 (36%)	175 (42%)	324 (39%)
Current smoker	205 (49%)	180 (43%)	385 (46%)
Ex-smoker	61 (15%)	61 (15%)	122 (15%)
Mean BMI (SD), kg/m ²	21.2 (3.4)	21.0 (3.8)	21.1 (3.6)
Anaemia			
Yes	37 (9%)	41 (10%)	78 (9%)
No	378 (91%)	375 (90%)	753 (91%)
Type 2 diabetes			
Yes	17 (4%)	18 (4%)	35 (4%)
No	398 (96%)	398 (96%)	796 (96%)
Comorbidities†			
Yes	120 (29%)	114 (27%)	234 (28%)
No	295 (71%)	302 (73%)	597 (72%)

Data are n (%) unless otherwise stated. Percentages may not add to 100% due to rounding. NIMR=National Institute for Medical Research. SATVI=South African Tuberculosis Vaccine Initiative. UCTLI=University of Cape Town Lung Institute.

*Race was reported by site investigator as self-reported by the participant.

†Comorbidities defined as a history of chronic or intermittent diseases, medical conditions, or symptoms judged to influence the primary endpoint as assessed by the medical expert (ÁHB) of the trial before unmasking, including type 2 diabetes.

Table 1: Baseline characteristics of trial participants (intention-to-treat analysis set)

In per-protocol analyses, recurrent tuberculosis occurred in 19 of 350 participants in the H56:IC31 group (12 relapses, four reinfections, and three indeterminate); and in ten of 347 participants in the placebo group (six relapses, three reinfections, and one indeterminate), equivalent to tuberculosis recurrence risk per 100 person-years of follow-up of 5.9% (95% CI 3.6–9.2) and 3.1% (1.5–5.6), respectively. Vaccine efficacy was calculated as –107.4% [–365.6 to 1.7]; $p=0.056$; figure 2D). Protocol-defined efficacy analyses in different analysis sets and exploratory analyses showed a consistent direction of effect (appendix 2 p 12).

In prespecified subgroup analyses, increased hazard for tuberculosis recurrence was observed for H56:IC31 versus placebo in individuals older than 35 years, individuals without anaemia, and individuals who received two doses, but interaction tests were not significant (figure 3).

In the immunogenicity analysis set at day 70, 2 weeks after the planned second dose, H56:IC31 had significantly increased the frequencies of H56-specific CD4 T cells expressing IFN- γ , TNF, IL-2, or IL-17 in vaccinees (median percentage of CD4 T cells, 0.35% [IQR 0.19–0.57]) compared with placebo (0.11% [0.09–0.23]; $p<0.0001$). H56-specific IgG responses were significantly higher in H56:IC31 recipients (median arbitrary units per mL, 6.84

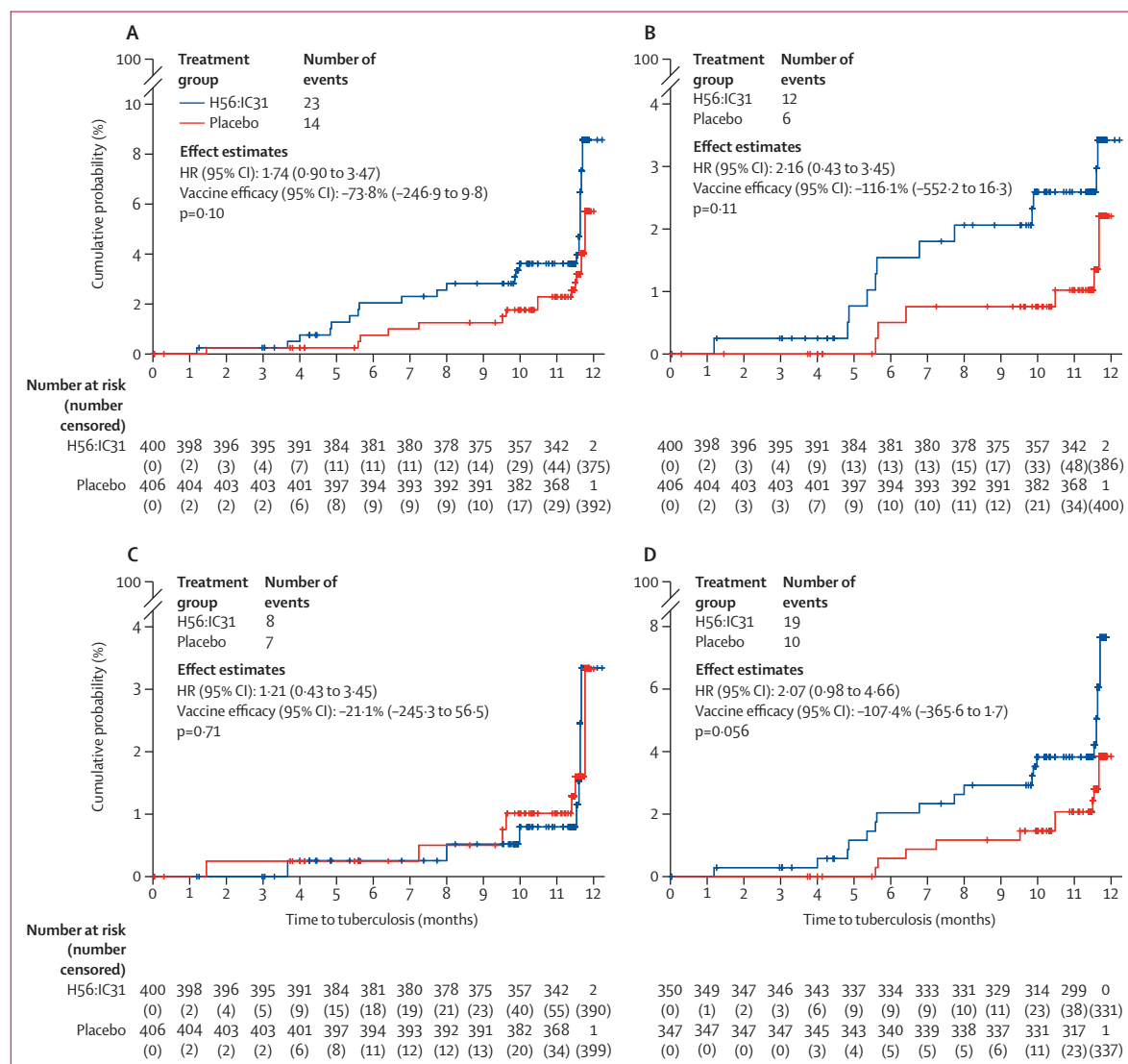


Figure 2: Time-to-first event analysis for primary and secondary efficacy endpoints

(A) Cumulative incidence of the primary trial endpoint of culture-confirmed recurrent tuberculosis (relapses, reinfections, or indeterminate) from day 70 (mITT analysis). (B) Cumulative incidence of the secondary endpoint of relapses with the same bacteria strain from day 70 (mITT analysis). (C) Cumulative incidence of the secondary endpoint of reinfections by a different bacteria strain from day 70 (mITT analysis). (D) Cumulative incidence of the primary trial endpoint of culture-confirmed recurrent tuberculosis (relapses, reinfections, or indeterminate) from day 70 (per-protocol analysis). p values for vaccine efficacy were derived from two-sided likelihood ratio tests. HR=hazard ratio. mITT=modified intention to treat.

[IQR 1.64–32.8]) than in placebo recipients (1.94 [1.05–3.86]; $p<0.0001$). In the placebo group, concentrations of anti-H56 IgG at baseline were significantly higher than at day 70 (figure 4).

In the safety analysis set (831 participants), a treatment-emergent adverse event was reported by 300 (72%) of 415 participants in the H56:IC31 group and by 267 (64%) of 416 participants in the placebo group (table 2). No treatment-related serious adverse events were reported. A greater proportion of H56:IC31 recipients had mild-to-moderate injection site reactions (165 [40%] of 415 vs 78 [19%] of 416) and myalgia (91 [22%] vs 57 [14%]) than placebo recipients (table 2,

appendix 2 p 13). Most treatment-related adverse events were mild to moderate and transient (appendix 2 p 14). Four (1%) participants in the H56:IC31 group and seven (2%) participants in the placebo group discontinued vaccination or the trial, or both, because of a treatment-emergent adverse event.

During the trial, 26 participants had a serious treatment-emergent adverse event unrelated to the study intervention (14 [3%] in the H56:IC31 group and 12 [3%] in the placebo group). Eight participants died: two (<1%) in the H56:IC31 group and six (1%) in the placebo group. Reported causes of death were pneumonia ($n=1$) and gunshot wound ($n=1$) in the H56:IC31 group, and cor

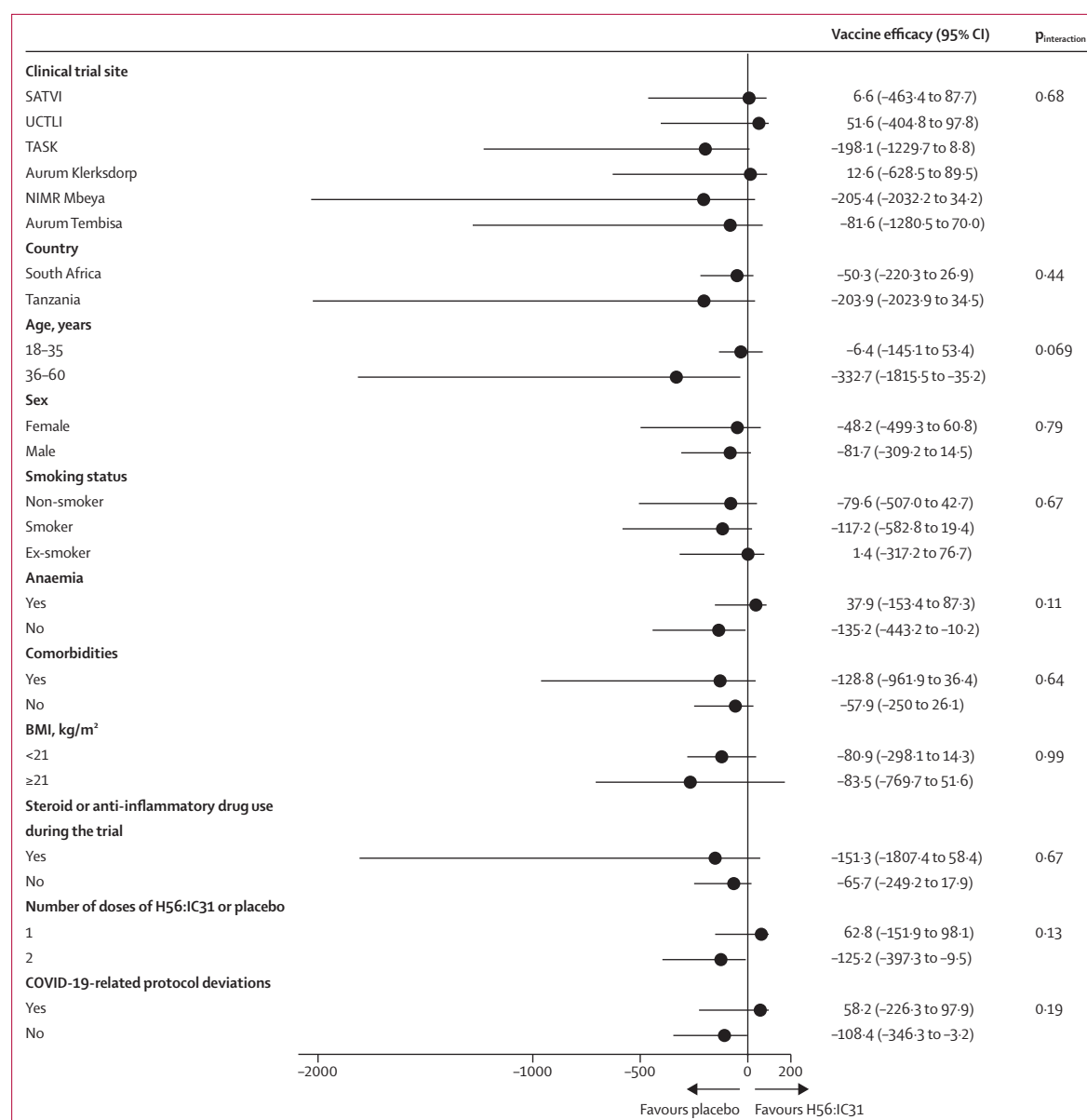


Figure 3: Vaccine efficacy estimates for H56:IC31 versus placebo across prespecified participant subgroups (mITT analysis set)

NIMR=National Institute for Medical Research. SATVI=South African Tuberculosis Vaccine Initiative. UCTLI=University of Cape Town Lung Institute.

pulmonale (n=1), death (natural cause; n=1), road traffic accident (n=1), gunshot wound (n=2), and craniocerebral injury (n=1) in the placebo group.

In the safety cohort, no notable group differences were observed and no marked changes from visit 2 to visits 4 and 6 were noted for any biochemistry or haematology laboratory safety tests (appendix 2 pp 15–22).

Discussion

To the best of our knowledge, this is the first report of a randomised, placebo-controlled clinical trial assessing the efficacy of a tuberculosis vaccine candidate to prevent disease recurrence after 5 months of treatment and confirmed cure by programmatic definition. WGS or tNGS of paired sputum samples provided an opportunity to differentiate relapses from reinfections. H56:IC31 was

well tolerated and immunogenic, but did not reduce the risk of recurrent tuberculosis as the primary endpoint. We observed an excess of nine cases of recurrence (including six additional relapses) among H56:IC31 vaccinees compared with placebo recipients, however the trial was not sufficiently powered to draw definitive conclusions about vaccine efficacy against the secondary endpoint of relapse (same strain of bacteria).

Recurrent tuberculosis imposes a substantial burden in endemic regions. Overall, 2–8% of patients completing tuberculosis treatment will have a recurrence in the following year.⁸ In countries with a high force of infection such as South Africa, recurrent tuberculosis amounts to 40% of all reported cases in some districts.⁷ Relapses are the most common mechanism of recurrence in the first year following an episode of tuberculosis.^{6,8} Reinfections by exogenous strains usually occur after the first year depending, among other factors, on the local prevalence of tuberculosis and HIV.^{6,9} Given that more than 10 million people globally acquired tuberculosis in 2023,¹ recurrence of disease after successful treatment and apparent cure remains a large, unmet medical need, which could be addressed by an effective vaccine.

Given the high incidence of tuberculosis recurrence in endemic regions,^{6–9} POR trials can be designed as smaller, shorter, and more cost-effective studies than prevention-of-disease (known as POD) trials in the general population. Furthermore, the endpoint in POR trials for tuberculosis is active tuberculosis disease, which is more clinically relevant than IFN- γ release assay test results that are used as endpoints in prevention-of-infection trials.²² However, POR trials might set a high performance bar for testing new tuberculosis vaccine candidates because immune mechanisms required to control mycobacteria, which are perturbed during active pulmonary disease, are not fully restored after successful treatment.^{23,24} The results of an ongoing trial in India and Bangladesh testing the recombinant BCG vaccine VPM1002 (NCT03152903) with a similar POR design will be helpful to clarify whether participants treated for active tuberculosis can be protected against relapse by vaccination.

In the current study, the observed excess cases of relapses among H56:IC31 vaccinees, although not statistically significant, raises a number of questions: (1) were the H56:IC31 and placebo groups comparable in severity of tuberculosis disease and pre-vaccination risk of relapse? (2) Were participants with a particular combination of a-priori risk factors in the H56:IC31 group at increased risk of relapse? And (3) can the observed higher rate of recurrence in the H56:IC31 group be explained by antigen-mediated or adjuvant-mediated mechanisms?

Prevaccination differences in relapse risk at the start or end of treatment could explain the excess relapse cases observed in the H56:IC31 group. Relapse risk increases with more severe disease,²⁵ characterised by pulmonary

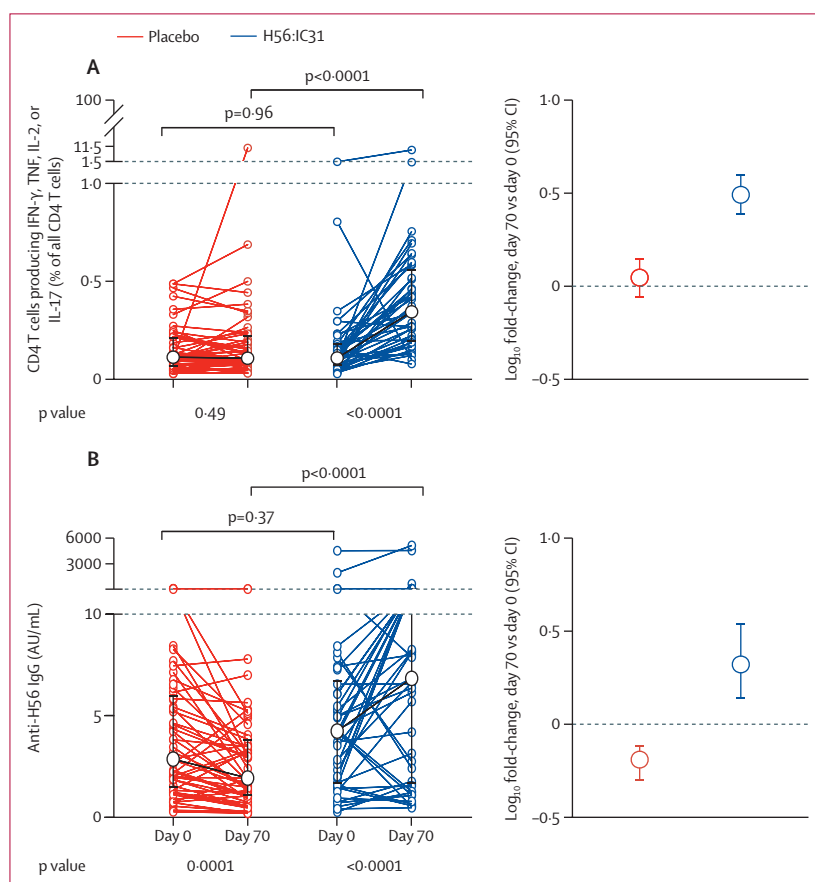


Figure 4: H56:IC31 immunogenicity (immunogenicity analysis set)

(A) H56-specific CD4-T-cell responses (40 H56:IC31 recipients and 48 placebo recipients). Antigen-specific CD4 T cells expressing any combination of IFN- γ , TNF, IL-2, or IL-17 after stimulation with recombinant H56 protein and measured by a whole-blood intracellular cytokine staining assay in participants receiving placebo or H56:IC31, expressed as percentage of CD4 T cells. (B) Concentrations of anti-H56 IgG (AU/mL) measured by ELISA (43 H56:IC31 recipients and 50 placebo recipients). Black hollow dots indicate medians, error bars the IQR. For fold-increase plots, the dots represent the median fold-change (day 70 vs day 0) and error bars represent 95% CI. p values denote comparisons between baseline (day 0) and day 70 (below the graph), or between the placebo and H56:IC31 group responses (above the graph), calculated with Wilcoxon signed-rank test and Mann-Whitney U test, respectively. $p < 0.05$ was considered to indicate statistical significance. AU=arbitrary units. IFN- γ =interferon- γ . IL=interleukin. TNF=tumour necrosis factor.

cavitation and high bacillary load. Furthermore, in treatment-shortening trials, the strongest predictor of poor treatment outcome was treatment adherence below 90%.²¹ This trial was not primarily designed to investigate prevaccination risk factors for relapse or reinfection, and the definition of cure was aligned with the programmatic definition (negative sputum smear after 5 months of treatment). Despite randomisation, there might have been unknown imbalances in disease severity at baseline, treatment adherence, or sputum smear-negative, but culture-positive, disease, towards the end of treatment, which in turn would have made the study groups dissimilar with respect to risk for relapse.

We speculate that mechanisms for the observed excess of nine cases of recurrence (including six relapses) among H56:IC31 vaccinees in our study might include: (1) vaccine-non-specific mechanisms, whereby H56:IC31 might have amplified an existing detrimental immune response in participants with tuberculosis towards the end of treatment, with the implication that other tuberculosis vaccine candidates might show similar results in trials with a POR design; and (2) vaccine-specific mechanisms, whereby the antigen H56 or the adjuvant IC31 might have induced a detrimental *de novo* immune response, with the implication that other tuberculosis vaccine candidates might show different results in trials with a POR design.

Vaccine-induced immune responses could have promoted the development of relapse by reactivation of endogenous bacteria. One hypothetical mechanism relates to the expansion of CD4 T cells activated by antigens co-shared by the vaccine and the bacteria. For instance, an increased risk of developing tuberculosis is observed with CD4 T-cell recovery following antiretroviral therapy initiation in advanced HIV,^{26,27} and after immunotherapy with programmed cell death protein-1 blockade in patients with cancer.²⁸ In our previous study in participants vaccinated at the end of tuberculosis treatment, H56:IC31 induced a CD4 T-cell response for Ag85B and ESAT-6 which persisted up to 6 months after vaccination.¹⁸ It is also possible that specific H56 antigens or IC31 components contributed to immune-mediated mechanisms, including induction or expansion of potentially unfavorable humoral or T-cell responses such as T-helper-2 cells or regulatory T cells,^{29,30} which might facilitate a permissive environment for the viable bacteria persisting in granulomas or draining lymph nodes at the end of treatment.

Our study has some limitations. It was not designed or powered to show efficacy in separately preventing relapses or reinfections, which were secondary endpoints. Due to the 1-year follow-up period after the second dose, we are unable to assess tuberculosis risk and the dynamics of relapses versus reinfections over a longer period. Additionally, active case-finding of asymptomatic tuberculosis occurred only at the last study visit. Our

	H56:IC31 group (N=415)	Placebo group (N=416)
Any treatment-emergent adverse event	300 (72%); 1382	267 (64%); 1033
Solicited treatment-emergent adverse event*	239 (58%); 1034	195 (47%); 706
Treatment-emergent adverse event of special interest†	0	2 (<1%); 2‡
Serious treatment-emergent adverse event	14 (3%); 16	12 (3%); 12
Fatal treatment-emergent adverse event	2 (<1%); 2	6 (1%); 6
Treatment-related adverse event	242 (58%); 1074	190 (46%); 741
Treatment-related serious adverse event	0	0
Suspected unexpected serious adverse reaction to investigational medicinal product	0	0
Treatment-emergent adverse event leading to discontinuation of investigational medicinal product	1 (<1%); 1‡	4 (1%); 5§
Treatment-emergent adverse event leading to discontinuation of trial	3 (1%); 3¶	6 (1%); 6
Intensity of treatment-emergent adverse events		
Severe	15 (4%); 18	10 (2%); 10
Moderate	84 (20%); 165	93 (22%); 163
Mild	278 (67%); 1199	234 (56%); 860
Outcome of treatment-emergent adverse events at trial end**		
Fatal	2 (<1%); 2	6 (1%); 6
Not recovered or not resolved	30 (7%); 36	29 (7%); 33
Recovered or resolved with sequelae	4 (1%); 4	3 (1%); 3
Recovered or resolved	293 (71%); 1340	253 (61%); 987
Unknown	0	4 (1%); 4
Treatment-emergent adverse events with a group frequency ≥5%		
General disorders and administration site conditions	216 (52%); 664	168 (40%); 389
Fatigue	112 (27%); 160	104 (25%); 155
Injection site pain	141 (34%); 212	57 (14%); 72
Injection site erythema	77 (19%); 98	36 (9%); 45
Injection site swelling	73 (18%); 101	23 (6%); 27
Pyrexia	26 (6%); 31	23 (6%); 28
Musculoskeletal and connective tissue disorders	118 (28%); 245	97 (23%); 188
Myalgia	91 (22%); 123	57 (14%); 79
Arthralgia	68 (16%); 92	62 (15%); 83
Nervous system disorders	104 (25%); 150	109 (26%); 167
Headache	100 (24%); 136	105 (25%); 158
Gastrointestinal disorders	76 (18%); 99	77 (19%); 102
Nausea	53 (13%); 63	47 (11%); 59
Infections and infestations	63 (15%); 77	60 (14%); 76
Upper respiratory tract infection	18 (4%); 18	25 (6%); 27

Data are number of participants (%); number of events. Adverse events were classified by system organ class and preferred term (Medical Dictionary for Regulatory Activities version 21.1). *Solicited adverse events were events occurring within 14 days after a vaccination and of the following types: injection site reactions (redness, swelling, and tenderness or pain) and systemic adverse events (pyrexia, arthralgia, myalgia, fatigue, headache, rash, chills, and nausea). Adverse events of special interest were a subset of adverse events that included autoimmune diseases and other systemic disorders of interest which could potentially have an autoimmune origin. †Facial paralysis (n=1) and erythema multiforme (n=1). ‡Seizure. §Gunshot wound (n=2), craniocerebral injury (n=1), pruritus generalised (n=1), and angioedema (n=1; pruritus generalised and angioedema occurred in the same participant). ¶Pneumonia (fatal; n=1), gunshot wound (fatal; n=1), and hospitalisation due to pulmonary tuberculosis (n=1). ||Gunshot wound (n=2), craniocerebral injury (n=1), cor pulmonale (n=1), death (natural cause; n=1), and road traffic accident (n=1); all fatal events. For adverse events leading to discontinuation of investigational medicinal product and adverse events leading to discontinuation of the trial, the same participant could be counted in the two categories. **Participants withdrawn from the study within 6 months after the last product administration were contacted at least 6 months after the last administration for recording of serious adverse events and adverse events of special interest.

Table 2: Treatment-emergent adverse events (safety analysis set)

results might have been different if spontaneous or induced sputum samples were collected for bacterial testing at more frequent intervals. We are also unable to assess whether non-specific effects of vaccination might promote development of symptoms, which would lead to the detection of otherwise asymptomatic tuberculosis. We did not use imaging to characterise pulmonary lesions, which would have detected asymptomatic cases and indicated disease severity. Regarding the influence of treatment adherence on recurrence risk, we did not collect information on adherence, and thus this aspect was not assessed. We also acknowledge that culture of expectorated sputum might have missed viable, but non-culturable bacilli and (culture-negative) pathology detectable by PET-CT. Potential methodological limitations of our study include measurement bias in the mITT and ITT population-based estimates, inherent selection bias in hazard ratios in the per-protocol and mITT population-based estimates, and selection bias due to missing outcome data. Instrumental analysis and G-estimation could have been helpful to address these biases.

In this first report of a randomised controlled trial of a novel tuberculosis vaccine candidate with a POR design, vaccination with H56:IC31 after 5 months of tuberculosis treatment was safe, well tolerated, and immunogenic, but did not reduce the risk of the primary endpoint, recurrent tuberculosis, in the year following last vaccination. Epidemiological and immunological analyses, including of possible vaccine-induced immune correlates of risk, are currently being performed by our group and are planned for future publication, to understand the observed trend towards increased risk of recurrence driven by the secondary endpoint, relapse with the same strain of bacteria, in the H56:IC31 group.

Contributors

DT, TJS, EN, MR, PA, RD, GC, IS, AHD, and MH conceived the study. RD, GC, IS, AHD, and MH were co-principal investigators and were responsible for participant recruitment and supervision of the clinical work. JS, LW, CI, PS, BN, CH, and TC were clinical trial site investigators. AHB and RM were responsible for overall consortium oversight and drafted the manuscript with MH. KTR performed sample size calculations and contributed to protocol development. RES and CHH accessed and verified the data and performed the statistical analyses. GKW and CGJ supervised H56:IC31 manufacture according to European Medicines Agency good manufacturing practice. TJS, EN, and AG designed the cell-mediated immune experiments and analysed the results. MPK qualified and did the serum IgG ELISA analyses. AMC did sequencing procedures and interpreted the genomic data. All authors contributed to data interpretation, critically revised the manuscript, and approved the final version. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol.

Declaration of interests

PA was a co-inventor on a patent on the H56 antigen as a TB vaccine. This patent was assigned to Statens Serum Institut and has now been abandoned. PA and RM are also co-inventors of patents covering the use of other tuberculosis vaccines. All rights for these patents have been assigned to Statens Serum Institut, a Danish not-for-profit institute under the auspices of the Danish Ministry of Health. All other authors declare no competing interests.

Data sharing

De-identified participant safety and immunogenicity data have been uploaded to a central repository (<https://zenodo.org/records/11610286>) and may be shared upon request. Research proposals should be submitted to the corresponding author (albs@ssi.dk). The proposal will be discussed for methodologically soundness by the POR TB study group. The research proposal's scientific relevance, design, statistical power, feasibility, and overlap with already published data will be assessed. Upon completion of the review, feedback will be provided to the proposer or proposers. In some circumstances, a revision of the concept might be requested. If the concept is approved for implementation, a writing group will be established consisting of the proposers and representatives from the POR TB study group. Study documents such as the trial protocol and informed consent forms may also be shared upon written request to the corresponding author, per the same process. WGS and tNGS data are available in the Sequence Read Archive of the National Center for Biotechnology Information as FASTQ files, under study accession number PRJNA1171160.

Acknowledgments

We thank all trial participants and clinical trial site personnel involved in the conduct of this trial. We thank Virginia Batignani, Marco Rossi, and Alberto Trovato for their support with sequencing and phenotypic testing. We also thank members of the scientific advisory committee, Ann Ginsberg, and independent ethics consultant Marzelle Haskins for their contribution to the present study. This study was funded by the European and Developing Countries Clinical Trials Partnership (EDCTP2) supported by the EU (grant number: RIA2016V-1631, POR TB consortium). Additional funding to support completion of the trial was provided by the Statens Serum Institut, The Aurum Institute, and the South African Tuberculosis Vaccine Initiative. Statens Serum Institut and IAVI were co-sponsors of the trial.

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